Andrew J. Miller

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EDUCATION

University of Colorado Boulder — Boulder, CO
Bachelor of Science in Mechanical Engineering; Minor in Business (Leeds School of Business) — *Expected May 2028*
GPA: 4.0/4.0 | Junior standing by credit hours
Relevant Coursework: Mechanics of Solids, Data Analysis & Experimental Methods, Circuits & Electronics, CAD & Fabrication

CERTIFICATIONS

Certified SolidWorks Associate (CSWA): Dassault Systèmes, Dec 2024
DeepLearning.AI: ChatGPT Prompt Engineering for Developers, Jun 2026; AI Agents in LangGraph, Jun 2026

TECHNICAL SKILLS

CAD & Design: SolidWorks (CSWA), Sheet Metal, assemblies, engineering drawings, tolerancing, GD&T, DFM, reverse engineering
Fabrication: Machining, welding, soldering, laser cutting, FDM 3D printing (Bambu Studio), mechanical assembly
Testing & Data: Experiment design, sensor characterization and calibration, ground-truth modeling, V&V, root cause analysis
Programming: MATLAB, Python, Arduino (C++), Processing, I2C and serial communication, LLM APIs, LangGraph
Electronics: Embedded systems integration, ultrasonic / time-of-flight / IR sensors, servo control, electronics debugging

EXPERIENCE

University of Colorado Boulder, College of Engineering — Boulder, CO
Teaching Assistant, Engineering Projects (GEEN 1400) — *Jan 2026 – Present*
- Mentor first-year engineering teams through the design, build, and test of projects that solve real-world problems by integrating mechanical and electrical systems
- Debug student hardware and Arduino code in labs and office hours, coaching teams through structured troubleshooting of sensors, wiring, and firmware

Fiberlok Technologies — Fort Collins, CO
Manufacturing Technician — *Summers 2023 – 2025*
- Improved department process efficiency and productivity by 30% through standardized procedures, workflow optimization, and waste reduction; improvement was adopted into formal training
- Operated the screens department solo, running photopolymer screen equipment from chemical mixing through photopolymerization

Hueber Industries — Fort Collins, CO
Fabrication Intern — *Jun 2024 – Aug 2024*
- Fabricated metal structures from engineering drawings using machining and welding, holding tight tolerances for dimensional accuracy and build quality
- Resolved a 250 lb railing installation failure through field diagnosis and shop rework

PROJECTS

ARC-001: Ultrasonic vs. Time-of-Flight Sensor Characterization | *Personal Project* — Summer 2026 – Present
- Designed and ran a servo-swept test rig comparing an HC-SR04 ultrasonic sensor and a VL53L1X time-of-flight (ToF) sensor against a six-object ground-truth scene of flat, rotated, metallic, and porous targets
- Quantified a 4.0× angular resolution advantage for the ToF sensor (13.7° vs. 54.6°) on a 13-gap resolution ladder at 300 mm
- Found and corrected 12 firmware and pipeline defects before final collection, then logged 9,720 rows across 20 sweeps with 0 rejected

Bus Capacity Tracking System | *CU Boulder, GEEN 1400 Engineering Projects* — Spring 2025
- Served as design and manufacturing lead for an Arduino-driven IR sensing system that tracked and displayed bus occupancy
- Reached 98% detection accuracy, reducing detection error below 2% through structured verification testing and iterative electrical and mechanical design changes
- Won the section best project award from department leaders

Reverse-Engineered Honeywell Fan | *CU Boulder, MCEN 1025 CAD & Fabrication* — Fall 2024
- Rebuilt the rotating system as a five-part SolidWorks assembly with roughly 130 features across the two most complex housings, emphasizing dimensional accuracy and tolerance awareness

Internship Search Automation Pipeline | *Personal Project* — Sep 2026 – Present
- Built a deterministic Python search tool with adapters for nine job-board platforms and narrowed 4,546 fetched postings to 14 relevant engineering internships

ENGINEERING TEAMS & ACTIVITIES

Formula SAE (FSAE) CU Boulder Racing Team — Boulder, CO
Team Member, EV Powertrain & Chassis Subsystems — *Sep 2026 – Present*
- Contributing to the electric-vehicle powertrain and chassis subsystems of the team's Formula SAE competition car